Inclusivity Evaluation of LLM generated code

Mandatory questions are marked with a star (*)

Survey Statement

We are conducting this survey as part of a study on evaluating a security mechanism (for inclusivity) generated using Large Language Models (LLMs) as a joint research project between LUT University in Finland and RHUL, University of London. The following questions guide the research.

Effectiveness: How effectively can LLMs generate code that implements security and inclusivity requirements, particularly for neurodivergent users?

Influencing Factors: What factors (e.g., prompt design, model choice, and specification level) influence the quality of LLM-generated code with respect to security and inclusivity?

Evaluation: How does LLM-generated code perform when reviewed by humans and LLMs in terms of identifying security vulnerabilities and inclusivity gaps?

The study included six cases across two user studies, with prompts varying from minimal to highly detailed inclusivity specifications; however, static security requirements were specified across all cases.

Please evaluate the code from each case for inclusivity. The survey will take approximately 30 minutes to complete.

We sincerely appreciate your time and participation.

Please note that no personally identifiable information will be collected during this survey.

Participation consent *

☐ I agree to participate in the survey

Current role

Number of years of experience

_____

Case I: No Inclusivity Specification

User Story
Helena is a 67-year-old patient with chronic heart disease. She calls the hospital to book an appointment to review her medication dosage. Due to changed privacy conditions, she has to accept the updated conditions by logging in to her user account for hospital authorities to be able to book an appointment for her. The authentication to her personal healthcare account requires a strong password, which she has forgotten; therefore, she needs to reset the password to be able to access the account and accept the privacy statement.

Please use ONE of the following links to access the application:
- https://llm-auth-s1-case1.japskua.workers.dev/?judge=1
- https://llm-auth-s1-case1.japskua.workers.dev/?judge=11

Email: helena@example.test

Inclusivity Evaluation criteria

Scale (1-5)
1: Strongly disagree, 2: Disagree, 3: Neutral, 4: Agree, 5: Strongly Agree

Scoring: All statements are positively keyed, such that higher scores indicate greater inclusivity. No reverse coding is required.

Attention *

	1	2	3	4	5
User interface is clean and does not have moving images or pop-ups. *	○	○	○	○	○
Important items clearly stand out. *	○	○	○	○	○
It is easy for the user to find what they need to do on the screen. *	○	○	○	○	○

Memory *

	1	2	3	4	5
The user does not need to remember many rules or steps. *	○	○	○	○	○
Options like Password manager, autofill, or “remember me” are present *	○	○	○	○	○
The user can easily recognize saved or familiar items. *	○	○	○	○	○

Comprehension *

	1	2	3	4	5
Text is short, clear, and easy to read. *	○	○	○	○	○
Error messages describe the problem and the way to fix it. *	○	○	○	○	○
Examples show what to type or how to do a step. *	○	○	○	○	○

Decision Making *

	1	2	3	4	5
Choices (like password, code, or face login) are clear. *	○	○	○	○	○
The user is not provided with too many choices at one time. *	○	○	○	○	○
The system clearly informs the outcome of each action. *	○	○	○	○	○

Learning *

	1	2	3	4	5
The user gets required help or short tips for new steps. *	○	○	○	○	○
The user can easily try again if she makes a mistake. *	○	○	○	○	○

	1	2	3	4	5
The design stays consistent so that the user can learn it over time. *	○	○	○	○	○

Case II: Moderate Inclusivity Specification

User Story

Helena is a 67-year-old patient with chronic heart disease and cognitive condition ADHD. She calls the hospital to book an appointment to review her medication dosage. Due to changed privacy conditions, she has to accept the updated conditions by logging in to her user account for hospital authorities to be able to book an appointment for her. The authentication to her personal healthcare account requires a strong password, which she has forgotten; therefore, she needs to reset the password to be able to access the account and accept the privacy statement.

Please use ONE of the following links to access the application:
- https://llm-auth-s1-case2.japskua.workers.dev/?judge=2
-https://llm-auth-s1-case2.japskua.workers.dev/?judge=22

Email: helena@example.com OR helena67

Inclusivity Evaluation criteria

Scale (1-5)
1: Strongly disagree, 2: Disagree, 3: Neutral, 4: Agree, 5: Strongly Agree

Attention *

	1	2	3	4	5
User interface is clean and does not have moving images or pop-ups. *	○	○	○	○	○
Important items clearly stand out. *	○	○	○	○	○
It is easy for the user to find what they need to do on the screen. *	○	○	○	○	○

Memory *

	1	2	3	4	5
The user does not need to remember many rules or steps. *	○	○	○	○	○
Options like Password manager, autofill, or “remember me” are present *	○	○	○	○	○
The user can easily recognize saved or familiar items. *	○	○	○	○	○

Comprehension *

	1	2	3	4	5
Text is short, clear, and easy to read. *	○	○	○	○	○
Error messages describe the problem and the way to fix it. *	○	○	○	○	○
Examples show what to type or how to do a step. *	○	○	○	○	○

Decision Making *

	1	2	3	4	5
Choices (like password, code, or face login) are clear. *	○	○	○	○	○
The user is not provided with too many choices at one time. *	○	○	○	○	○
The system clearly informs the outcome of each action. *	○	○	○	○	○

Learning *

	1	2	3	4	5
The user gets required help or short tips for new steps. *	○	○	○	○	○
The user can easily try again if she makes a mistake. *	○	○	○	○	○

	1	2	3	4	5
The design stays consistent so that the user can learn it over time. *	○	○	○	○	○

Case III: Detailed Inclusivity Specification

User Story

Helena is a 67-year-old patient with chronic heart disease and cognitive condition ADHD. She calls the hospital to book an appointment to review her medication dosage. Due to changed privacy conditions, she has to accept the updated conditions by logging in to her user account for hospital authorities to be able to book an appointment for her. The authentication to her personal healthcare account requires a strong password, which she has forgotten; therefore, she needs to reset the password to be able to access the account and accept the privacy statement.

Inclusivity description: People with ADHD can lose track of multi-step tasks, get easily distracted, and feel overwhelmed by unclear instructions or time pressure. The login and password reset flow must be forgiving, clear, and low-stress. Avoid dense instructions, timeouts, or unexpected page changes. Support memory and focus by showing visible progress and offering reminders of the next step.

Additional requirements:

• Keep the process simple and structured with clear, step-by-step guidance.
• Use straightforward and consistent language without unnecessary information.
• Provide clear feedback at each step so the user knows what has happened and what to do next.
• Allow enough time for actions without rushing or session timeouts.
• Reduce visual and cognitive distractions in the interface.
• Ensure the user can pause and return to the task without losing progress.
• Make help options easy to find and access at any stage.

Please use ONE of the following links to access the application:

- https://llm-auth-s1-case3.japskua.workers.dev/?judge=3
- https://llm-auth-s1-case3.japskua.workers.dev/?judge=33

Email: helena.patient@examplehospital.org

Inclusivity Evaluation criteria

Scale (1-5)
1: Strongly disagree, 2: Disagree, 3: Neutral, 4: Agree, 5: Strongly Agree

Attention *

	1	2	3	4	5
User interface is clean and does not have moving images or pop-ups. *	○	○	○	○	○
Important items clearly stand out. *	○	○	○	○	○
It is easy for the user to find what they need to do on the screen. *	○	○	○	○	○

Memory *

	1	2	3	4	5
The user does not need to remember many rules or steps. *	○	○	○	○	○
Options like Password manager, autofill, or “remember me” are present *	○	○	○	○	○
The user can easily recognize saved or familiar items. *	○	○	○	○	○

Comprehension *

	1	2	3	4	5
Text is short, clear, and easy to read. *	○	○	○	○	○
Error messages describe the problem and the way to fix it. *	○	○	○	○	○
Examples show what to type or how to do a step. *	○	○	○	○	○

Decision Making *

	1	2	3	4	5
Choices (like password, code, or face login) are clear. *	○	○	○	○	○
The user is not provided with too many choices at one time. *	○	○	○	○	○

	1	2	3	4	5
The system clearly informs the outcome of each action. *	○	○	○	○	○

Learning *

	1	2	3	4	5
The user gets required help or short tips for new steps. *	○	○	○	○	○
The user can easily try again if she makes a mistake. *	○	○	○	○	○
The design stays consistent so that the user can learn it over time. *	○	○	○	○	○

Case I: No Inclusivity Specification

User Story II

Marcus is a 54-year-old self-employed contractor who has just opened an account with an online bank. He manages his account through the bank's mobile web application, a responsive website he opens in his phone's browser. Following a regulatory update, the bank requires all customers to enrol in multi-factor authentication (MFA) before authorising payments above a threshold. Marcus signs in, verifies his identity, sets up a time-based one-time passcode (OTP) authenticator, and securely stores a set of backup recovery codes.

Please use ONE of the following links to access the application:

- https://llm-auth-s2-case1.japskua.workers.dev/?judge=4
- https://llm-auth-s2-case1.japskua.workers.dev/?judge=44

Test credentials are provided on the application page.

Inclusivity Evaluation criteria

Scale (1-5)
1: Strongly disagree, 2: Disagree, 3: Neutral, 4: Agree, 5: Strongly Agree

Scoring: All statements are positively keyed, such that higher scores indicate greater inclusivity. No reverse coding is required.

Attention *

	1	2	3	4	5
User interface is clean and does not have moving images or pop-ups. *	○	○	○	○	○
Important items clearly stand out. *	○	○	○	○	○
It is easy for the user to find what they need to do on the screen. *	○	○	○	○	○

Memory *

	1	2	3	4	5
The user does not need to remember many rules or steps. *	○	○	○	○	○
The system provides options such as copy-to-clipboard or QR codes so the user does not have to manually enter long codes. *	○	○	○	○	○
The user can easily recognise the information and options needed to complete each step. *	○	○	○	○	○

Comprehension *

	1	2	3	4	5
Text is short, clear, and easy to read. *	○	○	○	○	○
Error messages describe the problem and the way to fix it. *	○	○	○	○	○
Examples show what to type or how to do a step. *	○	○	○	○	○

Decision Making *

	1	2	3	4	5
Choices (like password, code, or face login) are clear. *	○	○	○	○	○
The user is not provided with too many choices at one time. *	○	○	○	○	○
The system clearly informs the outcome of each action. *	○	○	○	○	○

Learning *

	1	2	3	4	5
The user gets required help or short tips for new steps. *	○	○	○	○	○
The user can easily try again if they makes a mistake. *	○	○	○	○	○
The design stays consistent so that the user can learn it over time. *	○	○	○	○	○

Case II: Moderate Inclusivity Specification

User Story II

Marcus is a 54-year-old self-employed contractor who has just opened an account with an online bank. He manages his account through the bank's mobile web application, a responsive website he opens in his phone's browser. Following a regulatory update, the bank requires all customers to enrol in multi-factor authentication (MFA) before authorising payments above a threshold. Marcus signs in, verifies his identity, sets up a time-based one-time passcode (OTP) authenticator, and securely stores a set of backup recovery codes. Marcus has dyslexia.

Please use ONE of the following links to access the application:

- https://llm-auth-s2-case2.japskua.workers.dev/?judge=5
- https://llm-auth-s2-case2.japskua.workers.dev/?judge=55

Test credentials are provided on the application page.

Inclusivity Evaluation criteria

Scale (1-5)

1: Strongly disagree, 2: Disagree, 3: Neutral, 4: Agree, 5: Strongly Agree

Scoring: All statements are positively keyed, such that higher scores indicate greater inclusivity. No reverse coding is required.

Attention *

	1	2	3	4	5
User interface is clean and does not have moving images or pop-ups. *	○	○	○	○	○
Important items clearly stand out. *	○	○	○	○	○
It is easy for the user to find what they need to do on the screen. *	○	○	○	○	○

Memory *

	1	2	3	4	5
The user does not need to remember many rules or steps. *	○	○	○	○	○
The system provides options such as copy-to-clipboard or QR codes so the user does not have to manually enter long codes. *	○	○	○	○	○
The user can easily recognise the information and options needed to complete each step. *	○	○	○	○	○

Comprehension *

	1	2	3	4	5
Text is short, clear, and easy to read. *	○	○	○	○	○
Error messages describe the problem and the way to fix it. *	○	○	○	○	○
Examples show what to type or how to do a step. *	○	○	○	○	○

Decision Making *

	1	2	3	4	5
Choices (like password, code, or face login) are clear. *	○	○	○	○	○
The user is not provided with too many choices at one time. *	○	○	○	○	○
The system clearly informs the outcome of each action. *	○	○	○	○	○

Learning *

	1	2	3	4	5
The user gets required help or short tips for new steps. *	○	○	○	○	○
The user can easily try again if they makes a mistake. *	○	○	○	○	○
The design stays consistent so that the user can learn it over time. *	○	○	○	○	○

Case III: Detailed Inclusivity Specification

User Story II

Marcus is a 54-year-old self-employed contractor who has just opened an account with an online bank. He manages his account through the bank's mobile web application, a responsive website he opens in his phone's browser. Following a regulatory update, the bank requires all customers to enrol in multi-factor authentication (MFA) before authorising payments above a threshold. Marcus signs in, verifies his identity, sets up a time-based one-time passcode (OTP) authenticator, and securely stores a set of backup recovery codes. Marcus has dyslexia.

Description of the Condition:

People with dyslexia can find dense text, long character strings, and time-pressured reading difficult, and may process written information more slowly. The enrolment and verification flow must minimise reading load, avoid requiring manual transcription of long codes, and never impose reading-based time limits. Support comprehension through plain language, generous spacing, and pairing text with icons.

Additional Requirements:

- Use a clear, legible, dyslexia-friendly typeface with adequate letter and line spacing (legible at mobile viewport sizes); avoid italics and all-caps for instructions.
- Keep instructions short and in plain language; avoid dense paragraphs and jargon; show a short example of any expected input, such as email or code format.
- Keep the interface free of moving, flashing, or auto-updating elements and visual clutter; make the

current step and the primary action visually prominent so the user can find what to do at a glance.
- Offer copy-to-clipboard and QR-code options, and support browser autofill and password managers, so the user never has to manually transcribe secrets or long codes.
- Present one clear primary action per screen, minimise simultaneous choices, and after each action confirm plainly what happened and what the next step is.
- Keep layout, wording, and step order consistent and predictable throughout; make brief help or hints easy to find at every step; and let the user retry any step, or reveal, hide, and re-request codes, without penalty.
- Do not impose time limits on reading steps; allow generous time to enter codes.
- Provide clear, specific error messages that state the problem and the fix without blaming the user.

Please use ONE of the following links to access the application:
- https://llm-auth-s2-case3.japskua.workers.dev/?judge=6
- https://llm-auth-s2-case3.japskua.workers.dev/?judge=66

Test credentials are provided on the application page.

Inclusivity Evaluation criteria

Scale (1-5)
1: Strongly disagree, 2: Disagree, 3: Neutral, 4: Agree, 5: Strongly Agree

Scoring: All statements are positively keyed, such that higher scores indicate greater inclusivity. No reverse coding is required.

Attention *

	1	2	3	4	5
User interface is clean and does not have moving images or pop-ups. *	○	○	○	○	○
Important items clearly stand out. *	○	○	○	○	○
It is easy for the user to find what they need to do on the screen. *	○	○	○	○	○

Memory *

	1	2	3	4	5
The user does not need to remember many rules or steps. *	○	○	○	○	○

	1	2	3	4	5
The system provides options such as copy-to-clipboard or QR codes so the user does not have to manually enter long codes. *	○	○	○	○	○
The user can easily recognise the information and options needed to complete each step. *	○	○	○	○	○

Comprehension *

	1	2	3	4	5
Text is short, clear, and easy to read. *	○	○	○	○	○
Error messages describe the problem and the way to fix it. *	○	○	○	○	○
Examples show what to type or how to do a step. *	○	○	○	○	○

Decision Making *

	1	2	3	4	5
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The user is not provided with too many choices at one time. *	○	○	○	○	○
The system clearly informs the outcome of each action. *	○	○	○	○	○

Learning *

	1	2	3	4	5
The user gets required help or short tips for new steps. *	○	○	○	○	○
The user can easily try again if they makes a mistake. *	○	○	○	○	○

	1	2	3	4	5
The design stays consistent so that the user can learn it over time. *	○	○	○	○	○